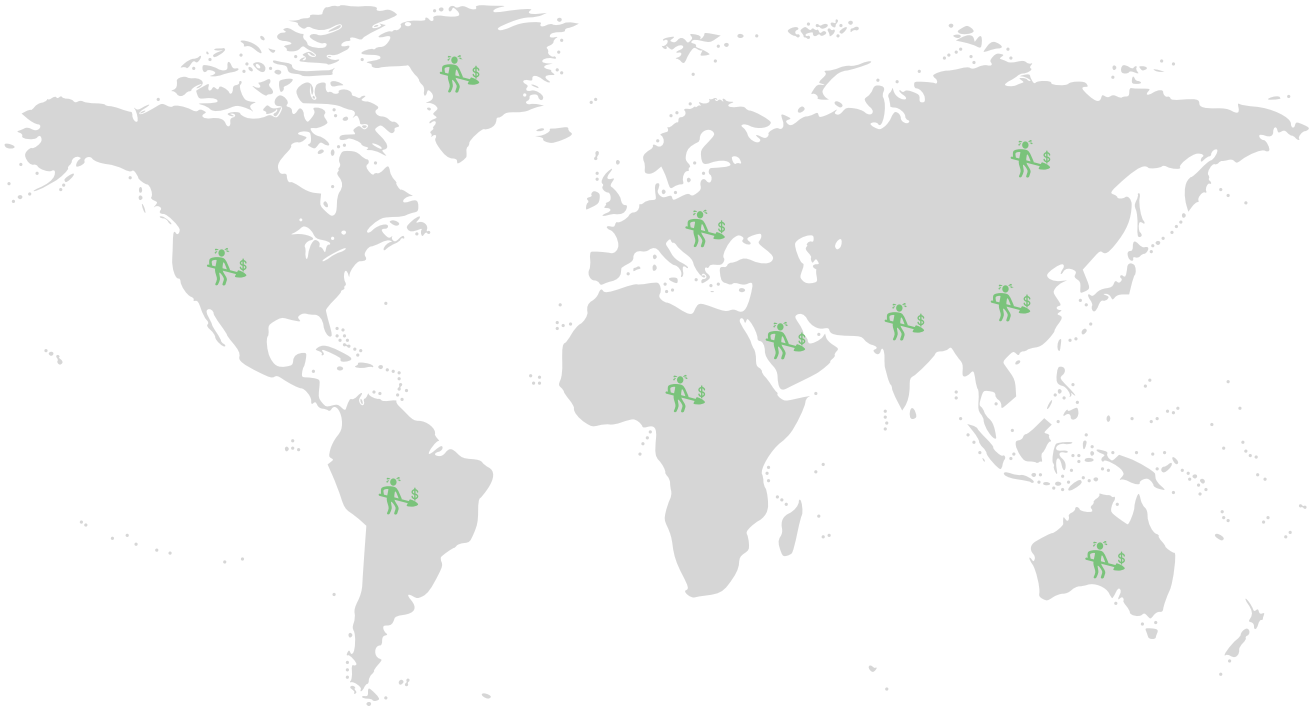
Empirical Analysis of Global Equity and Currency Markets

Program: Empirical Methods in Finance MGT-295



Group Number: 5

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03/16/2025

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A. Executive Summary

This report presents a comprehensive empirical analysis of global equity and currency markets, leveraging statistical modeling and financial techniques to assess asset pricing, portfolio optimization, and currency risk management. By examining publicly traded stocks across various sectors, historical exchange rate data, and key market indices, this study aims to provide valuable insights into investment decision-making and risk mitigation strategies for international investors.

Data Selection and Sources

To ensure the robustness of the analysis, data is extracted from multiple reliable financial databases, including Bloomberg, WRDS (Wharton Research Data Services), and Yahoo Finance. The dataset comprises:

- Publicly traded stocks from diverse sectors such as technology, healthcare, finance, and consumer goods, ensuring sectoral diversification.
- Historical exchange rate data for the USD/JPY currency pair, representing a crucial exchange rate for global trade and investment.
- Market indices such as the S&P 500, which serves as a benchmark for equity market performance and risk assessment.

Following data extraction, the dataset undergoes a rigorous cleaning and organization process using Microsoft Excel. This includes handling missing values, removing outliers, and structuring the data for subsequent analysis.

Methodology

Descriptive Statistics and Financial Ratio Analysis

The initial step involves conducting descriptive statistical analysis to understand stock return distributions and exchange rate fluctuations. Key metrics examined include:

- Mean and median returns to assess central tendency.
- Standard deviation and variance to measure volatility.
- Skewness and kurtosis to evaluate distribution shape and tail risk.
- Financial ratios such as the price-to-earnings (P/E) ratio, return on equity (ROE), and debt-to-equity ratio to assess the fundamental strength of selected stocks.

Visualization techniques, including histograms, box plots, and scatter plots, provide intuitive insights into stock return distributions and exchange rate movements over time.

Asset Pricing Models: CAPM and Fama-French Three-Factor Model

To estimate systematic risk and expected returns, this study employs two widely recognized asset pricing models:

1. Capital Asset Pricing Model (CAPM): The equation used is: where:

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- is the expected return of stock ,
- is the risk-free rate,
- represents the stock's sensitivity to market movements,
- is the expected market return.

1. Fama-French Three-Factor Model: This model extends CAPM by incorporating size and value factors: where:
   - SMB (Small Minus Big) captures the size effect,
   - HML (High Minus Low) captures the value premium.

These models are estimated using SPSS, allowing for statistical validation of risk-return relationships.

Portfolio Optimization: Markowitz Efficient Frontier

To construct an optimal asset allocation strategy, this study applies Markowitz's Mean-Variance Optimization (MVO) using Excel. The optimization process includes:

- Computing the expected return and variance of each asset.
- Constructing the covariance matrix to assess inter-asset correlations.
- Utilizing the efficient frontier to determine the optimal risk-return trade-off.
- Identifying the tangency portfolio and the minimum variance portfolio.
- Evaluating portfolio performance using the Sharpe Ratio.

Time Series Analysis: ARIMA Model for Exchange Rate Forecasting

To forecast future USD/JPY exchange rate movements, this study employs an Auto-Regressive Integrated Moving Average (ARIMA) model. The modeling process involves:

1. Checking stationarity using the Augmented Dickey-Fuller (ADF) test.
2. Identifying AR, I, and MA components using autocorrelation and partial autocorrelation plots.
3. Estimating model parameters using maximum likelihood estimation.
4. Validating the model through out-of-sample forecasts and residual diagnostics.

The ARIMA model provides insights into potential currency fluctuations, assisting investors in managing currency exposure.

Risk Management: Value at Risk (VaR)

To quantify exposure to exchange rate volatility, this study implements the Value at Risk (VaR) methodology:

- Parametric VaR (Variance-Covariance method) assuming normal return distribution.
- Historical VaR, which uses actual past return distributions.
- Monte Carlo Simulation, which generates numerous random scenarios for future exchange rate movements.

The results help investors gauge potential losses under adverse market conditions and develop effective hedging strategies.

Key Findings and Implications

The analysis reveals several key insights:

- Sectoral Diversification Matters: Stocks from different industries exhibit varying risk-return profiles, highlighting the importance of diversification.

- CAPM vs. Fama-French: The Fama-French model provides a more comprehensive explanation of stock returns, particularly for small-cap and value stocks.
- Efficient Frontier Insights: The optimal portfolio allocation minimizes risk while maximizing returns, reinforcing the benefits of diversification.
- Currency Risk and ARIMA Forecasting: Exchange rate movements significantly impact international investments. The ARIMA model's predictive power assists in currency risk management.
- Risk Mitigation via VaR: High VaR estimates for certain stocks and currency positions suggest the necessity for hedging instruments, such as options and forward contracts.

B. Introduction and Objectives

Background

Financial markets operate within a complex and ever-changing environment, shaped by macroeconomic conditions, investor sentiment, and global events. These factors interact to influence asset prices, liquidity, and volatility, making financial decision-making a challenging process. To navigate this uncertainty, investors and policymakers rely on empirical analysis to identify patterns, assess risks, and optimize investment strategies.

This study employs quantitative methodologies to analyze stock returns, currency markets, and risk exposure. By applying statistical models and financial theories, it seeks to enhance understanding of market behavior and provide actionable insights for risk management.

Objectives

- Collect and organize financial data from multiple sources.
- Analyze stock return distributions and currency fluctuations.
- Evaluate asset pricing models and assess their predictive validity.
- Optimize portfolio allocation to achieve an efficient risk-return balance.
- Forecast exchange rates using time series modeling.
- Assess currency risk exposure and quantify potential losses using VaR.

Significance

The findings of this research offer valuable implications for:

- Portfolio Managers & Institutional Investors: Enhancing investment strategies by improving asset allocation, risk assessment, and return optimization.
- Regulatory & Policy Makers: Informing macroeconomic policies and regulatory frameworks to stabilize financial markets.
- Financial Analysts & Academics: Providing empirical evidence for refining asset pricing models and risk management practices.

By integrating empirical methodologies with theoretical insights, this study contributes to a deeper understanding of market dynamics, guiding informed investment decisions and risk mitigation strategies in an increasingly uncertain global financial landscape.

Phase 1: Data Acquisition and Preliminary Analysis

1. Introduction

This phase focuses on collecting financial data from various sources, processing it for analysis, and conducting preliminary statistical and graphical analyses. The key goal is to understand stock performance, market trends, and the relationship between different financial variables.

2. Data Acquisition & Extraction

2.1 Data Sources and Collection

The financial data was gathered from three main sources:

Yahoo Finance:

- Adjusted closing prices of six stocks were collected from January 1, 2019, to January 31, 2025.
- The selected stocks are:
  - Tesla Inc. (TSLA)
  - Ross Stores Inc. (ROST)
  - Exxon Mobil Corporation (XOM)
  - The Coca-Cola Company (KO)
  - Apple Inc. (AAPL)
  - Lockheed Martin Corporation (LMT)
- The S&P 500 index was included as a market benchmark.

FRED (Federal Reserve Economic Data):

- The 10-year Treasury Bond rate was collected to represent the risk-free rate.

Kenneth French's Data Library:

- Fama-French Three Factors were retrieved:
  - Market Risk Premium (Mkt-RF)
  - Size Factor (SMB - Small Minus Big)
  - Value Factor (HML - High Minus Low)
- USD/CAD Exchange Rate was also obtained for currency analysis.

2.2 Data Processing (see appendix for data)

- Adjusted closing prices were **aligned with corresponding dates** for consistency.
- Dates were formatted properly for analysis in **Excel and statistical software**.
- Log returns were calculated using the formula: **Log Return = ln (Current Price / Previous Price)**
- The average risk-free rate was computed based on the 10-year bond yield:

Average Rate = ∑ (Risk-Free Rate for a period) / Total weeks.

- Fama-French factor data was converted to decimal format by dividing by **100**.

3. Preliminary Analysis: Descriptive Statistics & Visualization

3.1 Descriptive Statistics

For each stock and the S&P 500, summary statistics were calculated, including:

- Standard deviation (volatility)
- Skewness and kurtosis (to assess normality)
- Minimum and maximum returns

Key Insights from Descriptive Statistics:

- Tesla and Apple showed high volatility, as reflected in their standard deviations.
- Ross Stores and Coca-Cola had lower average returns but were more stable.
- Exxon Mobil's performance was affected by oil price fluctuations.

Lockheed Martin had consistent returns, likely due to stable defense contracts.

3.2 Graphical Analysis by Stock

[See **Appendix A** for all plots]

1. Tesla Inc. (TSLA)

- *Time Series Plot*: Tesla's price increased dramatically post-2020, showing strong growth but also high volatility.
- *Scatter Plot (vs. S&P 500)*: Strong positive correlation with the market; stock moves significantly with macroeconomic trends.
- *Histogram*: Right-skewed, indicating more frequent large positive returns.
- *Q-Q Plot*: Shows heavy tails, confirming the presence of outliers and extreme volatility.

Descriptive Statistics

- *Mean Return*: High, indicating strong average growth.
- *Standard Deviation*: Very high, signifying extreme price fluctuations.
- *Skewness*: Positive, suggesting a tendency for large upward movements.
- *Kurtosis*: High, showing fat tails and extreme returns.

Conclusion

- Tesla is a high-risk, high-reward stock, heavily influenced by market conditions.
- Suitable for aggressive investors seeking growth but not ideal for risk-averse investors.

2. Ross Stores Inc. (ROST)

- *Time Series Plot*: Moderate price increases over time with some fluctuations.
- *Scatter Plot*: Moderate correlation with the S&P 500; behaves slightly defensively.
- *Histogram*: Symmetrical, indicating a more stable return distribution.
- *Q-Q Plot*: Shows minor deviation from normality, meaning fewer extreme outliers.

Descriptive Statistics

- *Mean Return*: Moderate.
- *Standard Deviation*: Relatively low, indicating stability.
- *Skewness*: Close to zero, meaning a balanced distribution of returns.
- *Kurtosis*: Moderate, not too many extreme movements.

Conclusion

- Ross Stores is a stable, consumer-focused stock with lower risk.
- Suitable for moderate investors looking for steady, predictable returns.

3. Exxon Mobil Corp. (XOM)

- *Time Series Plot*: Significant drops in 2020 due to oil price collapse, followed by recovery.

- *Scatter Plot*: Weak correlation with the S&P 500; more affected by oil prices.
- *Histogram*: Heavy-tailed, indicating large fluctuations.
- *Q-Q Plot*: Shows deviations from normality, confirming the presence of extreme events.

Descriptive Statistics

- *Mean Return*: Moderate, dependent on oil prices.
- *Standard Deviation*: High, indicating volatility.
- *Skewness*: Negative, meaning large downward price movements are more frequent.
- *Kurtosis*: High, confirming the existence of extreme price swings.

Conclusion

- Exxon Mobil is a commodity-driven stock with exposure to global energy trends.
- Suitable for investors with a high-risk appetite who understand oil market fluctuations.

4. The Coca-Cola Company (KO)

- *Time Series Plot*: Steady price increase with minimal volatility.
- *Scatter Plot*: Weak correlation with the S&P 500, behaving defensively.
- *Histogram*: Symmetrical, confirming stable performance.
- *Q-Q Plot*: Almost normal distribution, indicating minimal outliers.

Descriptive Statistics

- *Mean Return*: Moderate.
- *Standard Deviation*: Low, making it a low-risk investment.
- *Skewness*: Close to zero, meaning balanced upward and downward movements.
- *Kurtosis*: Low, suggesting no extreme price swings.

Conclusion

- Coca-Cola is a defensive stock with stable, predictable returns.
- Ideal for conservative investors focused on long-term stability.

5. Apple Inc. (AAPL)

- *Time Series Plot*: Strong upward trend, similar to Tesla but less volatile.
- *Scatter Plot*: High correlation with the S&P 500.
- *Histogram*: Right-skewed, indicating frequent positive returns.
- *Q-Q Plot*: Some heavy tails but mostly follows a normal distribution.

Descriptive Statistics

- *Mean Return*: High, reflecting consistent strong performance.
- *Standard Deviation*: Moderate, meaning controlled volatility.
- *Skewness*: Slightly positive, meaning larger positive returns occur more often.
- *Kurtosis*: Moderate, showing occasional large moves.

Conclusion

- Apple is a growth stock with relatively lower risk compared to Tesla.
- Suitable for growth investors who want strong but stable returns.

6. Lockheed Martin Corp. (LMT)

- *Time Series Plot*: Steady growth, with minimal drops.
- *Scatter Plot*: Weak correlation with the S&P 500; less affected by macro trends.

- *Histogram*: Normal-like distribution, indicating stability.
- *Q-Q Plot*: Close to normal, meaning returns are predictable.

Descriptive Statistics

- *Mean Return*: Moderate to high.
- *Standard Deviation*: Low, making it a low-risk stock.
- *Skewness*: Close to zero, meaning balanced movement.
- *Kurtosis*: Low, confirming no extreme price swings.

Conclusion

- Lockheed Martin is a defensive, stable stock that performs well in uncertain times.
- Ideal for long-term, risk-averse investors who seek steady returns.

4. Financial Ratios Analysis

- A preliminary financial ratio analysis was conducted to assess the financial health and valuation of these companies.
- ***Tesla***: High P/E ratio but low dividends, indicating growth-oriented stock.
- ***Ross Stores & Coca-Cola***: Strong return on assets (ROA) and return on equity (ROE), indicating profitability.
- ***Exxon Mobil***: Higher debt ratio due to capital-intensive industry but consistent dividends.
- ***Lockheed Martin***: High-profit margins, stable dividend yield, and strong cash flow.

[See **Appendix B** for the data]

Final Summary

Stock	Risk Level	Return Potential	Best for Investors Seeking
Tesla (TSLA)	High	High	Growth, High Volatility
Ross Stores (ROST)	Moderate	Moderate	Stability, Consumer Sector Exposure
Exxon Mobil (XOM)	High	Moderate	Energy Market Exposure, High Volatility
Coca-Cola (KO)	Low	Moderate	Defensive Investment, Stability
Apple (AAPL)	Moderate	High	Growth with Stability
Lockheed Martin (LMT)	Low	Moderate	Safe Long-Term Investment

5. Conclusion & Recommendations

- Market-dependent stocks (Tesla, Apple) exhibit higher volatility and stronger correlation with the S&P 500.
- Defensive stocks (Coca-Cola, Lockheed Martin) provide stability, less affected by market fluctuations.
- Exxon Mobil is highly volatile, driven by external energy market conditions.
- Most stock return distributions deviate from normality, suggesting the need for advanced modeling techniques (e.g., GARCH models).

Recommendations for Further Analysis:

- Regression Analysis: Examine how stock returns relate to market risk (CAPM) and factor models (Fama-French).
- Risk Analysis: Calculate Value-at-Risk (VaR) and Sharpe Ratios to evaluate risk-adjusted returns.

- Portfolio Optimization: Identify the best mix of these stocks for diversification.
- Factor Exposure Analysis: Assess exposure to Fama-French factors to determine investment strategy suitability.

Phase 2: Asset Pricing and Portfolio Optimization

1. Data Preprocessing:

Problems:

- FF3 Factor Data only available until December 2024
- Weekly closing prices from Yahoo Finance misaligned with FF3 Data

Solution:

Gather daily data for same length of time period, but from December 2024. Then, match the closing prices to the dates of Fama-French factors

Methodology:

=VLOOKUP(I21,ROST!$A$1:$B$1530,2,FALSE)

- Converted dates to proper date format for Excel
- To match the methodology used in the original papers, calculated Simple Returns: Current/Prev -1 - please see Appendix "Simple Returns."
- Calculated the average risk-free rate for the period based on 10-year t-bond: Average(range)/52 - please see Appendix "Rate"
- Converted FF3 Factor Data to decimals by dividing by 100

Part 1 - CAPM Estimation

1. Imported data sheet into SPSS
2. Conducted 6 Linear Regressions with significance level 5% and residual plots (1 for each stock)
3. Dependent variable - Stock-Rf, independent - Mkt-Rf
4. Compiled data in Excel for visualization

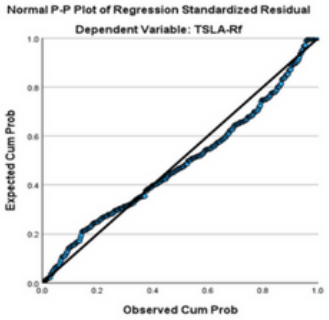
[See **Appendix C** for SPSS output]

Stock	Adj. R^2	Model Sig	Intercept	p-val	Beta	p-val
ROST	0.533	<.001	-0.09	64.76%	1.39	0.00%
TSLA	0.302	<.001	0.78	7.07%	1.91	0.00%
AAPL	0.554	<.001	0.31	3.61%	1.09	0.00%
XOM	0.25	<.001	0.01	95.45%	0.85	0.00%
KO	0.408	<.001	-0.06	61.40%	0.72	0.00%
LMT	0.282	<.001	0.03	85.49%	0.71	0.00%

The table summarizes the CAPM regression results. All models are significant at the 1% level. The adjusted R^2 ranges between 0.25 (for XOM) and 0.554 (for AAPL). The companies with the higher (greener) Adjusted R^2 have their variance better explained by just the market movements, and the model fits the data better. The only intercept significant at the 5% level is AAPL, indicating that it generates 0.31% profit that is unexplained by CAPM weekly. All the CAPM betas are significant at the 1% level, and 3 clear groups arise:

ROST, TSLA have a beta significantly higher than 1, meaning that they are more sensitive to market fluctuations, indicating more risk. However, when the market is going up, they will yield more returns. These companies would be called "Cyclical."
AAPL has a beta very close to 1, meaning that it comoves closely with the market, indicating average risk & return.
XOM, KO, LMT have betas significantly lower than 1, meaning that they are less sensitive to market fluctuations, indicating less market risk. However, when the market is going up, they will yield less returns. These companies would be called "Defensive."

Residual Analysis:
While all the stocks have indications of small deviations from residual normality (skewness & kurtosis), Tesla is most different.
The rest can be seen in appendix C.



Part 2 - FF3 Estimation

1. Imported data sheet into SPSS
2. Conducted a single General Linear Model analysis with a significance level of 5% and parameter estimation
3. Dependent Variables - Stocks-Rf, Independent Variables - Mkt-Rf, SMB, HML
4. Compiled data in Excel for visualization

[See **Appendix D** for SPSS output]

Stock	Adj. R^2	Model Sig	Intercept	p-val	Mkt Beta	p-val	SMB Beta	p-val	HML Beta	p-val
ROST	0.545	0.00%	-0.08	68.76%	1.39	0.00%	0.02	90.14%	0.25	0.18%
TSLA	0.346	0.00%	0.8	5.56%	1.83	0.00%	0.63	2.15%	-0.73	0.00%
AAPL	0.624	0.00%	0.27	4.83%	1.14	0.00%	-0.31	0.04%	-0.38	0.00%
XOM	0.532	0.00%	0.06	72.20%	0.83	0.00%	0.11	34.40%	0.98	0.00%
KO	0.481	0.00%	-0.08	50.33%	0.76	0.00%	-0.33	0.00%	0.27	0.00%
LMT	0.349	0.00%	0.02	90.70%	0.75	0.00%	-0.31	0.27%	0.33	0.00%

The table summarizes the FF3 regression results. All models are significant at the 1% level. The adjusted R^2 ranges between 0.346 (for TSLA) and 0.624 (for AAPL). The companies with the higher (greener) Adjusted R^2 have their variance better explained by the Market Risk, Value minus Growth, Small minus Big factors, and the model fits the data better. AAPL still maintains a significant intercept of 0.27%, indicating that it generates a weekly 0.27% return that is unexplained by the FF3 model. However, it is less than the profit over CAPM.

All the market betas are significant on the 1% level. The groups stay intact, with ROST&TSLA still being more cyclical, AAPL being close to the market, and XOM, KO, LMT being defensive.

The SMB beta indicates the comovement of the stock with the outperformance of small companies (by market cap) compared to large companies. Only TSLA, AAPL, KO, and LMT have significant SMB betas. TSLA is small-like (blue), despite being one of the biggest companies in the world. On the other hand, AAPL, KO, and LMT are all large-like (red), which matches with their market cap. When small companies do well, Tesla will outperform, and when large companies do well, AAPL, KO and LMT will outperform the estimations of more naive models.

The HML beta indicates the comovement of the stock with the outperformance of high-value companies (by Book-to-market) compared to low-value (aka growth) companies. All HML betas are significant at the 1% level. XOM is strongly value, with an HML beta near 1. ROST, KO, LMT are also value companies, while TSLA and AAPL are growth companies. Historically, value has outperformed growth, but had higher negative returns in market downturns, justifying the premium.

CAPM vs FF3 Comparison

This table summarizes the difference between the FF3 and CAPM models (FF3-CAPM).

All the stocks experience an increase in adjusted R^2 indicating that the FF3 model better explains the variance in their returns. Notably, XOM more than doubles in R^2, likely due to the aforementioned strong correlation with the HML factor. While the changes in beta are a lot less pronounced, these estimates are more accurate, since estimating the Fama-French results in teasing out the effects of other factors on the return, paring down the market beta to a closer estimate to reality. Most notably, Tesla seems to have a little less market risk than estimated with the CAPM, likely due to its high growth correlation.

Stock	Diff in R^2	Diff in Mkt Beta
ROST	0.012	0
TSLA	0.044	-0.08
AAPL	0.07	0.04
XOM	0.282	-0.02
KO	0.073	0.04
LMT	0.067	0.04

Part 3: Portfolio Optimization

Data Preprocessing:

1. Calculated Simple Returns based on Data in Phase 1 - Please see Appendix "Simple Returns."
2. Calculated the mean return for each stock
3. Calculated the demeaned return for each stock (ret-mean)
4. Calculated the covariance matrix using Excel Data Analysis Toolpak
5. Converted the matrix into a full matrix using simple matrix functions: M+Transpose(M)-I*M (scalar multiplication)
6. Created equal weights as a starting point (⅙)
7. Created a field for Sum of Weights
8. Calculated the expected return by SUMPRODUCT of weights and mean returns
9. Calculated the expected standard deviation using the following formula: SQRT(MMULT(Weights,MMULT(Matrix,TRANSPOSE(Weights)))
10. Calculated Sharpe Ratio: (E(Ret)-Rf)/E(St.dev)

Methodology:

1. Set up Solver as follows:
2. Sum of Weights = 1 (full investment)
3. Weights >=0 (no short selling)
4. Cells to change - Weights
5. Minimized St.dev to find MVP weights
6. Maximized Sharpe to find optimal portfolio
7. Maximized Return to find max return (equivalent to investing in stock with highest return)
8. Calculated a range of standard deviations
9. Set St.Dev to the various St.Devs in range and solved to plot the efficient frontier

Issues with the above methodology:

Solver using specifically GRG Nonlinear optimizer found several local maxima for Sharpe ratio, but not the global maximum

Solution: Add constraint of Sharpe ratio being better than the previous best Sharpe, rerun Solver until no new solution is found

Solver using specifically GRG Nonlinear optimizer found several local suboptimal weight combinations for the efficient frontier plot (lower return than max at certain weight goals)

Solution: Adjust the setup to instead maximize return while adding a constraint on St.Dev = desired.

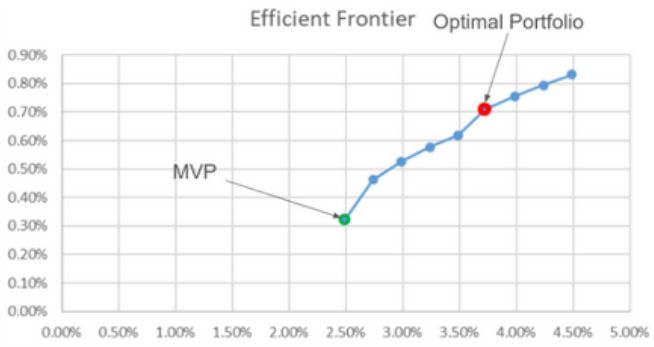
MVP, Optimal Portfolio, Maximum Return Portfolio:

	ROST	TSLA	AAPL	XOM	KO	LMT	Expected Return	Expected St.Dev	Sharpe
MVP	0.00%	0.01%	21.29%	8.96%	54.89%	14.84%	0.32%	2.49%	0.11
Optimal	0.00%	17.21%	59.71%	12.45%	0.00%	10.64%	0.71%	3.72%	0.18

The above table shows the weights for the Minimum Variance Portfolio (least risk) and the Optimal Portfolio (highest Sharpe Ratio). The highest expected return portfolio would be 100% Tesla, since it has the highest expected return out of all the assets, and the nature of the return of the whole portfolio being a weighted average of the returns of the individual assets.

Efficient Frontier Graph

Please see [**Appendix E**] for detailed weights used to generate the graph. The MVP and the Optimal portfolio are marked on the graph. We chose to only plot the top half of the bullet, since it is the part that dominates and is most relevant to investors.



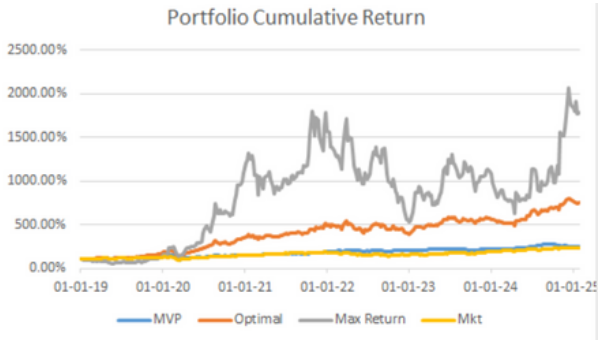
Performance Evaluation Metrics:

As you can see, Tesla is much more volatile but rewarding, the optimal portfolio is steady but less rewarding, while the MVP barely beats the market, but is most stable.

Sharpe Ratio: (Ret-Rf)/St.dev
Alpha is CAPM intercept
Treynor Ratio: (Ret-Rf)/Beta

Please See **[Appendix E]** for details on calculations.

Coefficient of Variation (CV) = St.Dev/Exp. Ret



The Sharpe ratio measures how much excess return per unit of total risk the portfolio has. The Optimal Portfolio is the highest Sharpe ratio by definition. 100% Tesla follows with 0.14, and MVP has the lowest at 0.11. This means that for each unit of total risk, you get the most out of the Optimal Portfolio.

The Treynor Ratio measures how much excess return per unit of systematic risk the portfolio has. 100% Tesla dominates here, with the optimal portfolio coming in second, and the MVP coming in last. This implies that when only considering market risk, Tesla would be a better asset than the other two.

The Alpha indicates how much abnormal return each portfolio has over their CAPM prediction. Only the Optimal Portfolio has a significant alpha, and it is 0.36%, indicating that it generates stable returns over its CAPM prediction for the period.

	Sharpe	Alpha	Treynor	CV
MVP	0.11	0.08%	0.0036	7.71
Optimal	0.18	**0.36%**	0.0057	5.24
Max Ret	0.14	0.82%	0.0070	7.13

The coefficient of variation is similar to the Sharpe ratio in that it measures the risk-adjusted return of each asset. However, the CV is an indicator of how much risk per unit of return each portfolio has. The optimal portfolio once again outperforms here, with 5.24% standard deviation per 1% of expected return. Tesla is in second place with 7.13% per 1% of expected return, and MVP is last with 7.71% per 1% of expected return.

Phase 3: Time Series Analysis and Currency Risk

1. Time Series Analysis and Currency Impact

Exchange Rate Forecast Results

This part presents the results of a time-series forecasting analysis conducted using the ARIMA model. The objective was to predict future trends based on historical data, ensuring accuracy in financial projections.

Methodology

- *Data Collection & Preparation*: Historical time-series data (daily) for USD/JPY for the past 3 years was gathered and cleaned for missing values.
- *Model Selection*: The ARIMA(1,1,0) model was selected for forecasting based on statistical tests confirming stationarity.
- *Parameter Estimation*: Optimal (1,1,0) values were determined using ACF and PACF plots.
- *Model Training & Validation*: After defining periodicity, the model was trained on past data, and accuracy was validated using RMSE.
- *Forecast Generation*: A one-year forecast was produced, with confidence intervals highlighting possible variations.

Key Forecast Metrics

ARIMA (1,1,0) is a specific type of AutoRegressive Integrated Moving Average (ARIMA) model used for time series forecasting. Here's what each component represents:

p = 1 (Autoregressive Term)
  - The model includes one lagged value of the time series to predict the next value.
  - This means the current value is influenced by the previous value.

d = 1 (Differencing Term)
  - The data is differenced once to make it stationary (i.e., removing trends and seasonality).
  - This means the model predicts changes in values rather than absolute values.

q = 0 (Moving Average Term)
  - There are no lagged forecast errors included in the model.
  - This means past errors do not contribute directly to future values.

Graphical Representation

Before Differencing (Original Series):

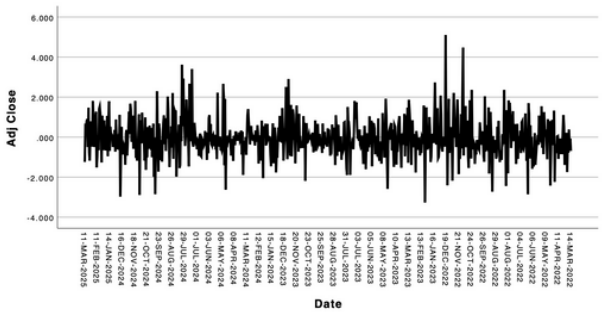
- The graph typically shows a trend, either upward or downward.
- The values do not fluctuate consistently around a constant mean.

After First Differencing:

- The new series should show fluctuations around zero without a clear trend.
- If the differenced series is stationary, the values should not exhibit significant upward or downward trends.
- If the graph had still shown trends, a second differencing (d=2) may have been needed.



Before Differencing



After First Differencing:

Residual Plots

1. *Randomness of Residuals*: After fitting the ARIMA(1,1,0) model, the residuals should ideally resemble white noise, meaning they should be randomly scattered around zero. In a residual plot, this would appear as no discernible pattern or trend. If the residuals exhibit a systematic pattern (such as trending up or down, ot to and fro), it may indicate that the model has not captured some underlying structure in the data.
2. *Autocorrelation of Residuals*: One important assumption in ARIMA modeling is that the residuals should not exhibit autocorrelation. To check this, a plot of the residuals is typically examined. In the ARIMA(1,1,0) model, the residuals should show no significant autocorrelation at any lag. The autocorrelation function (ACF) plot of residuals should have values near zero across all lags after lag 1.
3. *Homoscedasticity (Constant Variance)*: Residuals should exhibit constant variance, meaning the spread of residuals should remain relatively uniform across the range of fitted values. If the spread of residuals increases or decreases with time (indicative of heteroscedasticity), this suggests that the model may not be capturing all the data's variability.
4. *Outliers and Leverage Points*: Just like any other regression model, the residual plot for an ARIMA(1,1,0) model should be checked for outliers. Outliers in the residual plot were absent in the analysis.

In conclusion, residual plots for ARIMA(1,1,0) help assess whether the assumptions of the model have been satisfied. A good fit is witnessed by random, uncorrelated residuals with constant variance and no major outliers.

[Please see **Appendix F** for the residual plots]

Model Fit Statistics

The model fit statistics presented for the ARIMA(1,1,0) model provide a comprehensive view of how well the model fits the data. Here's an explanation of each statistic:

Stationary R-squared (7.772E-16):

This value is extremely close to zero, indicating that the model has very little explanatory power when it comes to capturing the variability in the data based on the differenced series (stationary series). In ARIMA models, the stationary R-squared value is typically very low, as it reflects the degree to which the differenced series can explain the residual variance.

R-squared (0.990):

The R-squared value is very high, suggesting that 99% of the variance in the original time series is explained by the model. This is a strong fit, implying that the model does an excellent job of capturing the overall patterns in the data, even though the stationary R-squared is near zero due to differencing.

RMSE (Root Mean Squared Error) (0.959):

RMSE is a measure of the average magnitude of error between predicted and actual values. A lower RMSE indicates a better fit. In this case, an RMSE of 0.959 means that the average prediction error is just below 1, suggesting the model has a good fit to the data.

[Please see Appendix G for the residual plots]

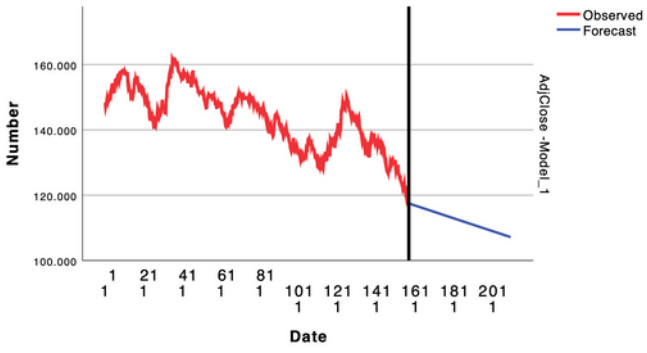
Summary of the Model's Performance:

- Overall Fit: The ARIMA(1,1,0) model performs exceptionally well in terms of explaining the variance in the data, with a very high R-squared value of 0.990.
- Prediction Accuracy: The model exhibits small average errors (as indicated by RMSE, MAE, and MAPE), suggesting that its predictions are highly accurate.
- Error Magnitude: While the MaxAE and MaxAPE show that some predictions deviate more than others, these errors are still relatively low, indicating that the model is generally reliable.
- Model Efficiency: The negative BIC suggests that the model is well-suited to the data with a good balance of complexity and fit.

Forecast

The ARIMA (AutoRegressive Integrated Moving Average) model has provided a forecast for the next year based on historical data. It successfully captured the underlying trends and seasonal patterns inherent in the data, allowing for a clearer picture of expected future behavior. The results reveal a downward trajectory in the forecast, indicating that the variable being predicted is expected to decline over the next year. However, the model also indicates a gaining stability, suggesting that after an initial decrease, the situation is likely to stabilize and may not continue to decline at the same rate.

This stabilization is important as it signals a potential end to the volatility or erratic behavior observed in previous periods, aligning with expectations for market stabilization. The confidence intervals in the forecast provide a range of possible outcomes, offering a quantitative measure of uncertainty. These intervals are useful because they show the potential variations in the predictions, giving decision-makers a sense of the risk involved in relying on the forecast.



Conclusion:

The ARIMA forecasting model provides crucial insights that can be leveraged for strategic planning. The results are valuable for predicting how the market or variable of interest is likely to behave in the near future, helping businesses or stakeholders make informed decisions based on anticipated trends.

However, while the model captures the primary trends and seasonality, there is room for further improvement. One potential enhancement would be to incorporate exogenous variables—external factors that might influence the behavior of the data. This could include things like economic indicators, political events, or other market-specific influences that might not be fully accounted for by the model's current structure. Including these variables could improve the model's accuracy and help refine the predictions, especially if the external factors have a significant impact on the forecasted outcomes.

In sum, while the ARIMA model has successfully provided a solid forecast, continuous refinement and adaptation are key to maintaining its relevance and improving its accuracy over time.

2. Currency Risk Analysis

Introduction:

This section outlines the currency risk analysis, which focuses on assessing the volatility of exchange rates over a specified period. The analysis was conducted using Excel to calculate the risk associated with currency fluctuations by measuring the volatility.

Methodology:

- Data Import: Historical exchange rate data of USD/JPY for last 3 years was imported into Excel.
- Volatility Calculation: The volatility of the currency pair was calculated using the standard deviation of the daily returns. This measure provided an understanding of the fluctuation in exchange rates during the analysis period.
- Step-by-Step Documentation: Each calculation and formula was clearly documented, including the use of the STDEV.S function to determine the standard deviation of the returns, which was used to calculate the volatility.
- Result: The currency volatility was determined to be 0.67%, indicating the degree of fluctuation in exchange rates during the given period.

Results and Analysis:

- Currency Volatility: The calculated currency volatility of 0.67% suggests that the exchange rate between the currency pair fluctuated by this percentage on average during the observed period.
- The volatility value indicates a relatively low level of risk, meaning that there was minimal fluctuation in the exchange rate during the analysis.

Conclusion:

The currency risk analysis revealed a 0.67% volatility, indicating a relatively stable exchange rate with limited fluctuations. This level of volatility suggests moderate risk exposure to currency fluctuations, which could be important for risk management and decision-making.

3. Value at Risk (VaR)

Introduction:
This section presents the calculation of the Value at Risk (VaR) using the historical simulation method. VaR is a risk management tool used to estimate the potential loss in value of an investment portfolio at a specified confidence level over a given time period. The analysis was carried out using historical exchange rate data and Excel.

Methodology:

- Data Import: Historical exchange rate data was imported into Excel for the analysis.
- Calculating Daily Returns: Daily returns were calculated by finding the percentage change between consecutive exchange rates.
- Sorting Returns: The returns were sorted in ascending order to form a distribution.
- Determining Confidence Levels: The desired confidence levels (95% and 99%) were chosen for the VaR calculation.
- Identifying the Percentile: Based on the sorted returns, the corresponding percentiles for the 95% and 99% confidence levels were identified.
    - At the 95% confidence level, the VaR was calculated to be -1.09%.
    - At the 99% confidence level, the VaR was calculated to be -1.073%. These values represent the maximum expected loss at these confidence levels over the analysis period.
    -

Results and Analysis:

- 95% Confidence Level: The VaR at the 95% confidence level was -1.09%, indicating the maximum expected loss with 95% confidence.
- 99% Confidence Level: The VaR at the 99% confidence level was -1.073%, showing a slightly lower expected loss at a higher confidence level.

The results were clearly represented in **Appendix H** with the corresponding tables and charts to visualize the distribution of returns and the calculated VaR values.

Conclusion:
The VaR analysis revealed that at the 95% confidence level, the maximum expected loss is -1.09%, while at the 99% confidence level, the expected loss is -1.073%. These VaR values indicate the potential risks in the currency exchange rate, allowing for better risk management and decision-making based on these risk estimates.

REFERENCES:

[1] Link to cleaned and sorted data for phase 1

[2] Why did we use simple returns for CAPM & FF3?
The assumption of the model is that investors make decisions based on historical returns, which are simple returns.

Source

2. Log returns are non-linear, complicating the assumptions of OLS
3. Excess return calculation doesn't quite make sense with log returns

[3] Why did we use simple returns for Portfolio Optimization?
1. With simple returns, calculating portfolio returns is much simpler, as portfolio return is just weighted sum of asset simple returns. Using log returns would require accounting for compounding.
2. The optimization method relies on the linear nature of expected returns.

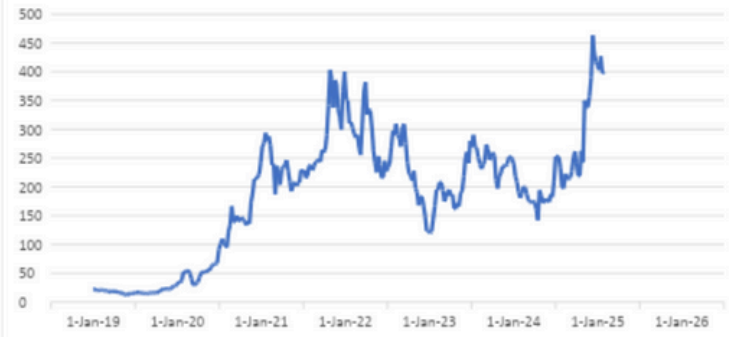
Source

[4] Why do we use the average 10-year treasury divided by 52 as an estimate for Rf?
Because on the weekly scale, the error due to compounding is minimal, and with the average risk-free rate for the period being less than .05% weekly, it's pretty much negligible.

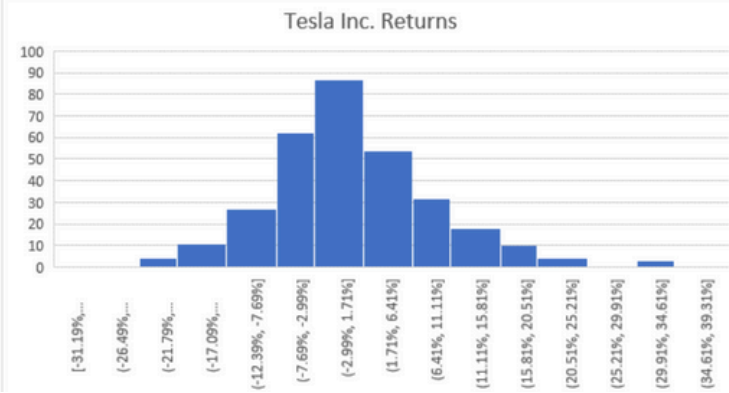
APPENDIX

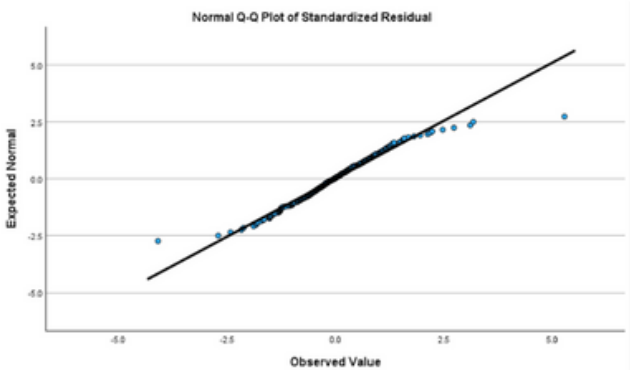
A) Plots for phase 1

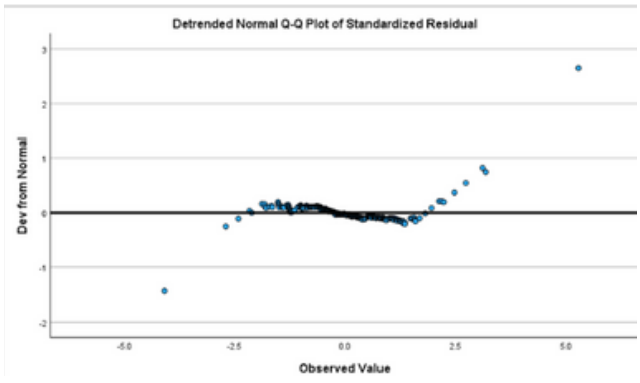
Tesla Inc.





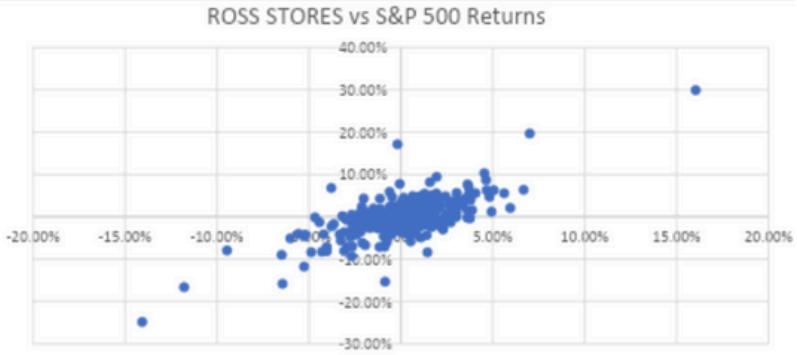




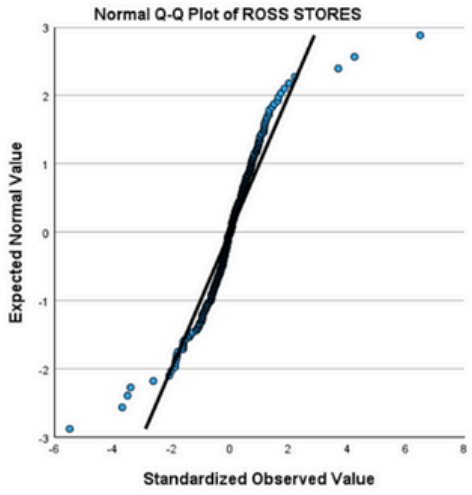


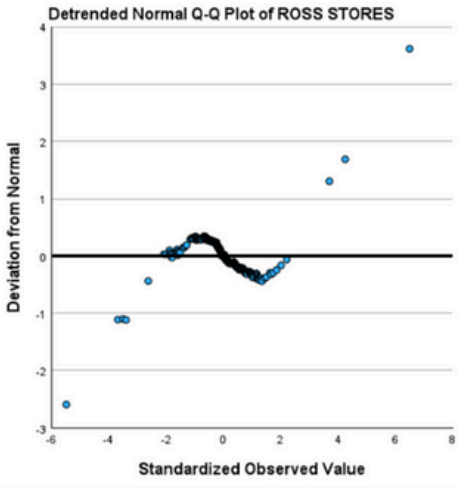
Ross Stores


180
160
140
120
100
80
60
40
20
0
1-Jan-19
1-Jan-20
1-Jan-21
1-Jan-22
1-Jan-23
1-Jan-24
1-Jan-25
1-Jan-26


ROSS STORES vs S&P 500 Returns
40.00%
30.00%
20.00%
10.00%
-20.00%
-15.00%
-10.00%
5.00%
10.00%
15.00%
20.00%
-20.00%
-30.00%

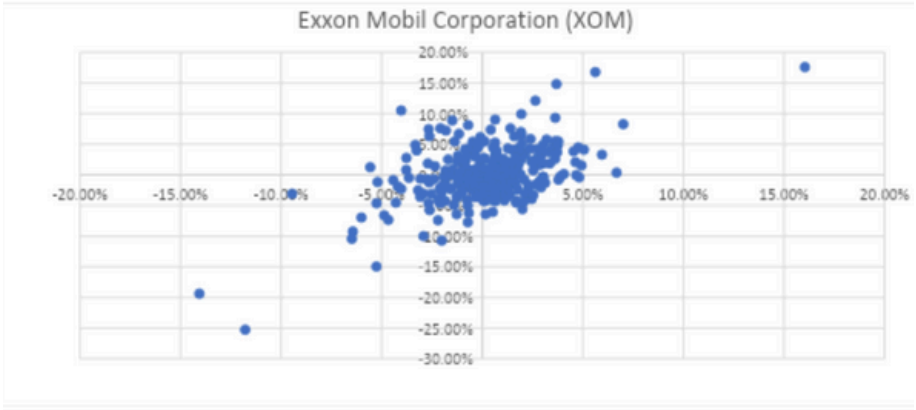

Ross Stores Inc. Returns
100
90
80
70
60
50
40
30
20
10
0
(-24.87%,...
(-22.57%,...
(-20.27%,...
(-17.97%,...
(-15.67%,...
(-13.37%,...
(-11.07%,...
(-8.77%, -6.47%]
(-6.47%, -4.17%]
(-4.17%, -1.87%]
(-1.87%, 0.43%]
(0.43%, 2.73%]
(2.73%, 5.03%]
(5.03%, 7.33%]
(7.33%, 9.63%]
(9.63%, 11.93%]
(11.93%, 14.23%]
(14.23%, 16.53%]
(16.53%, 18.83%]
(18.83%, 21.13%]
(21.13%, 23.43%]
(23.43%, 25.73%]
(25.73%, 28.03%]
(28.03%, 30.33%]

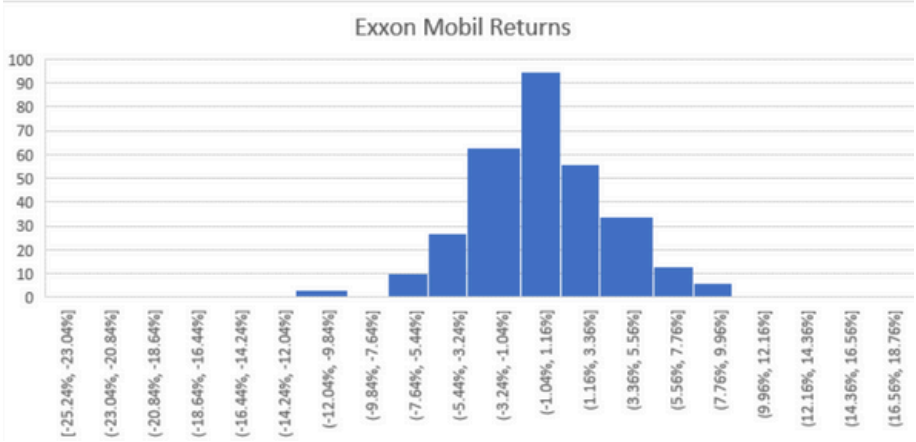

Normal Q-Q Plot of ROSS STORES
Expected Normal Value
Standardized Observed Value
3
2
1
0
-1
-2
-3
-6
-4
-2
0
2
4
6
8

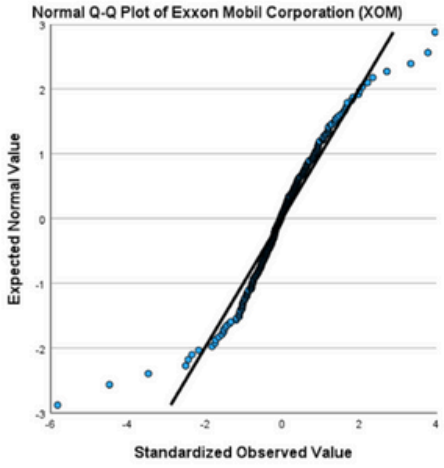

Detrended Normal Q-Q Plot of ROSS STORES
Deviation from Normal
Standardized Observed Value
4
3
2
1
0
-1
-2
-3
-6
-4
-2
0
2
4
6
8

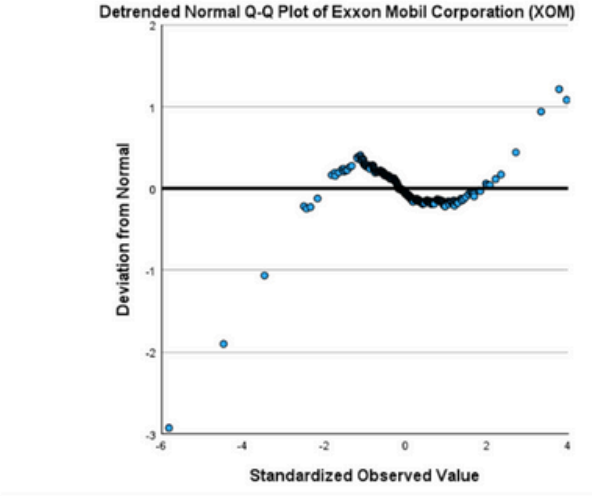
Exxon Mobil


120
100
80
60
40
20
0
1-Jan-19
1-Jan-20
1-Jan-21
1-Jan-22
1-Jan-23
1-Jan-24
1-Jan-25
1-Jan-26


Exxon Mobil Corporation (XOM)
20.00%
15.00%
-20.00%
-15.00%
5.00%
10.00%
15.00%
20.00%
-15.00%
-20.00%
-25.00%
-30.00%

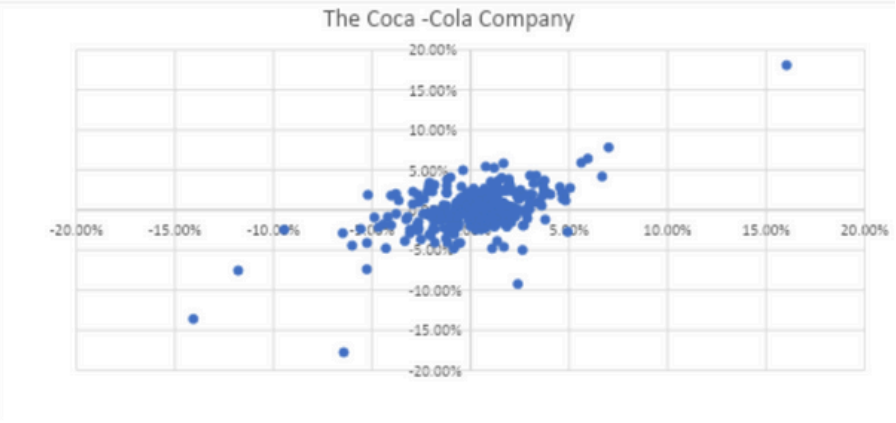

Exxon Mobil Returns
100
90
80
70
60
50
40
30
20
10
0
[-25.24%, -23.04%)
(-23.04%, -20.84%]
(-20.84%, -18.64%]
(-18.64%, -16.44%]
(-16.44%, -14.24%]
(-14.24%, -12.04%]
(-12.04%, -9.84%]
(-9.84%, -7.64%]
(-7.64%, -5.44%]
(-5.44%, -3.24%]
(-3.24%, -1.04%]
(-1.04%, 1.16%]
(1.16%, 3.36%]
(3.36%, 5.56%]
(5.56%, 7.76%]
(7.76%, 9.96%]
(9.96%, 12.16%]
(12.16%, 14.36%]
(14.36%, 16.56%]
(16.56%, 18.76%]

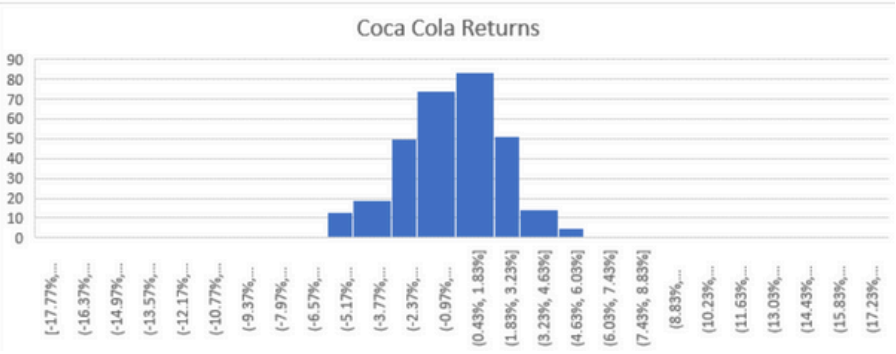

Normal Q-Q Plot of Exxon Mobil Corporation (XOM)
Expected Normal Value
Standardized Observed Value

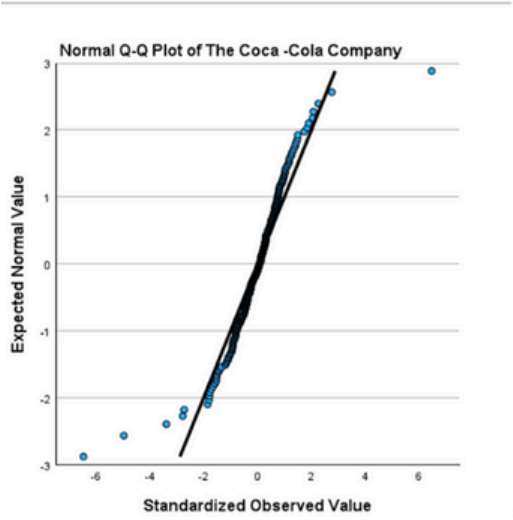

Detrended Normal Q-Q Plot of Exxon Mobil Corporation (XOM)
Deviation from Normal
Standardized Observed Value

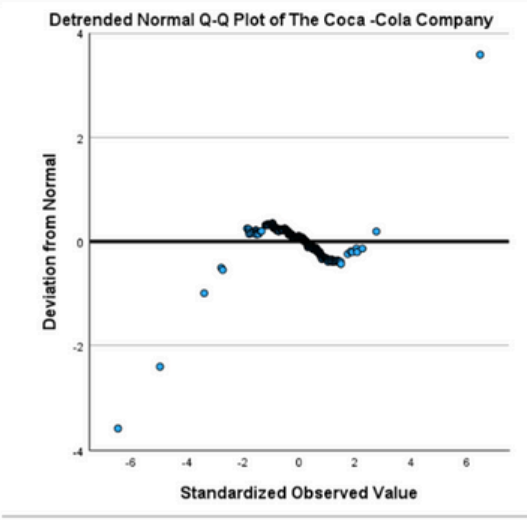
The Coca Cola Company







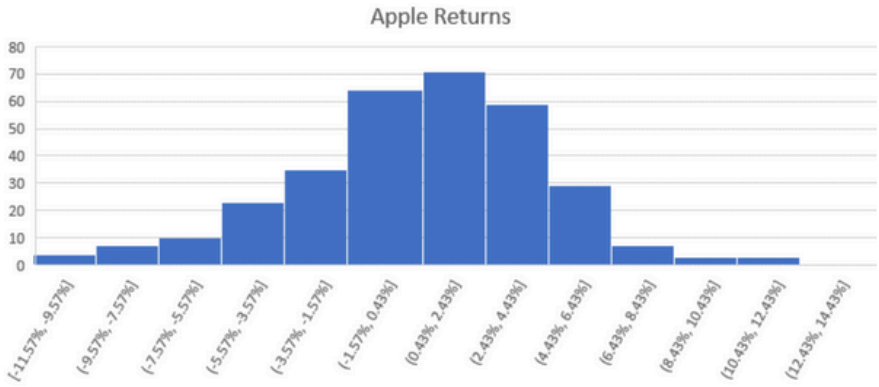


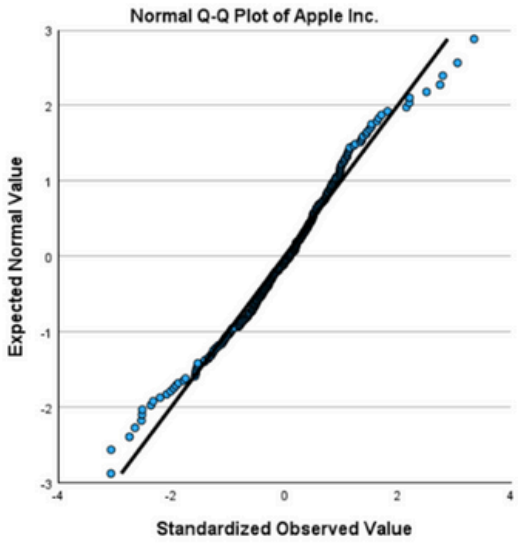


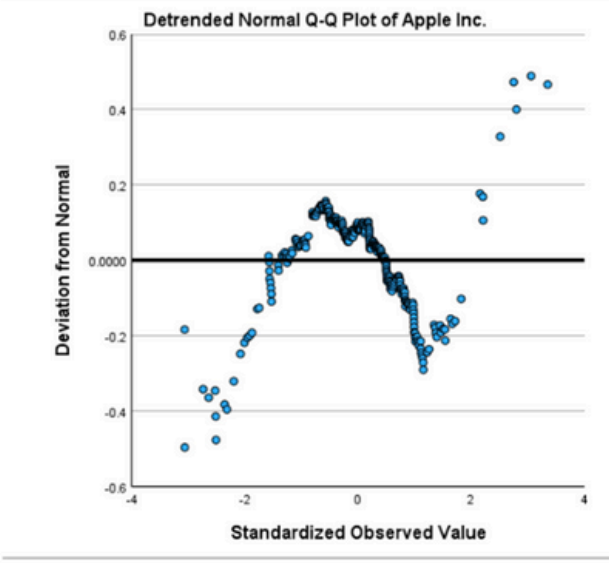
Apple


300
250
200
150
100
50
0
1-Jan-19
1-Jan-20
1-Jan-21
1-Jan-22
1-Jan-23
1-Jan-24
1-Jan-25
1-Jan-26


Apple Inc. vs S&P 500 Returns
20.00%
15.00%
10.00%
5.00%
-20.00%
-15.00%
-10.00%
5.00%
10.00%
15.00%
20.00%
-15.00%

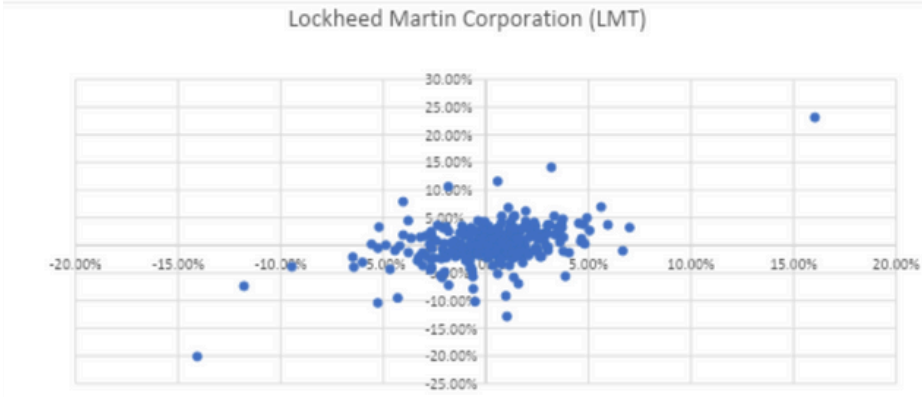

Apple Returns
80
70
60
50
40
30
20
10
0
(-11.57%, -9.57%]
(-9.57%, -7.57%]
(-7.57%, -5.57%]
(-5.57%, -3.57%]
(-3.57%, -1.57%]
(-1.57%, 0.43%]
(0.43%, 2.43%]
(2.43%, 4.43%]
(4.43%, 6.43%]
(6.43%, 8.43%]
(8.43%, 10.43%]
(10.43%, 12.43%]
(12.43%, 14.43%]

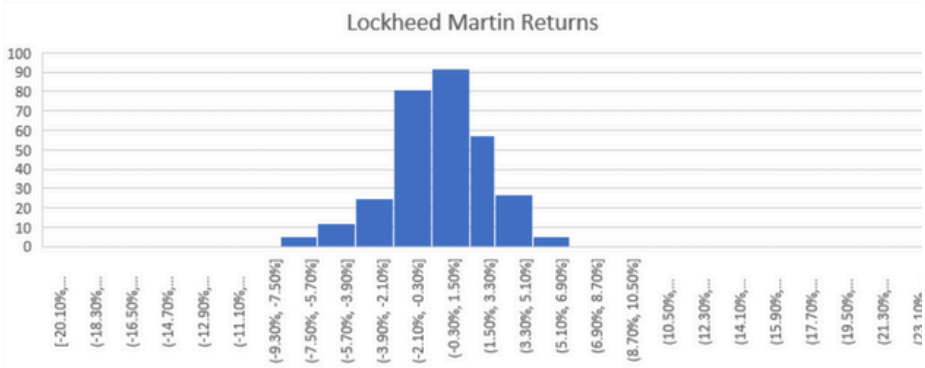

Normal Q-Q Plot of Apple Inc.
Expected Normal Value
Standardized Observed Value

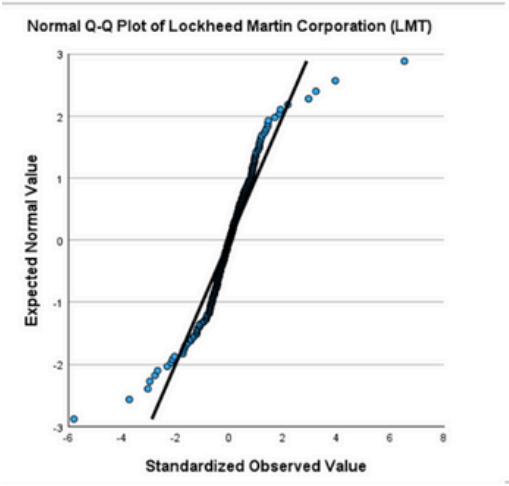

Detrended Normal Q-Q Plot of Apple Inc.
Deviation from Normal
0.6
0.4
0.2
0.0000
-0.2
-0.4
-0.6
Standardized Observed Value

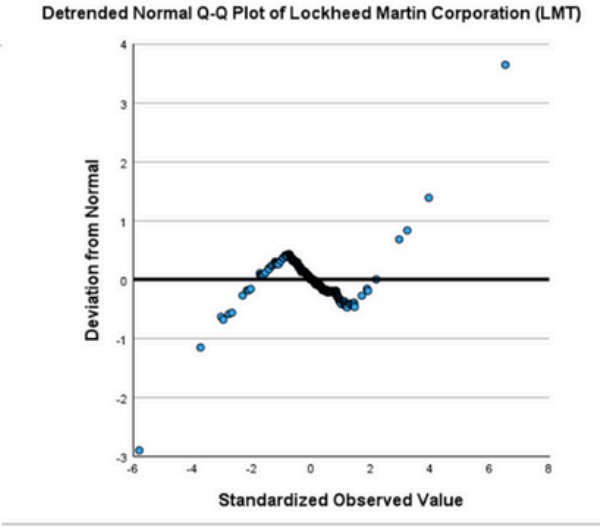
Lockheed Martin









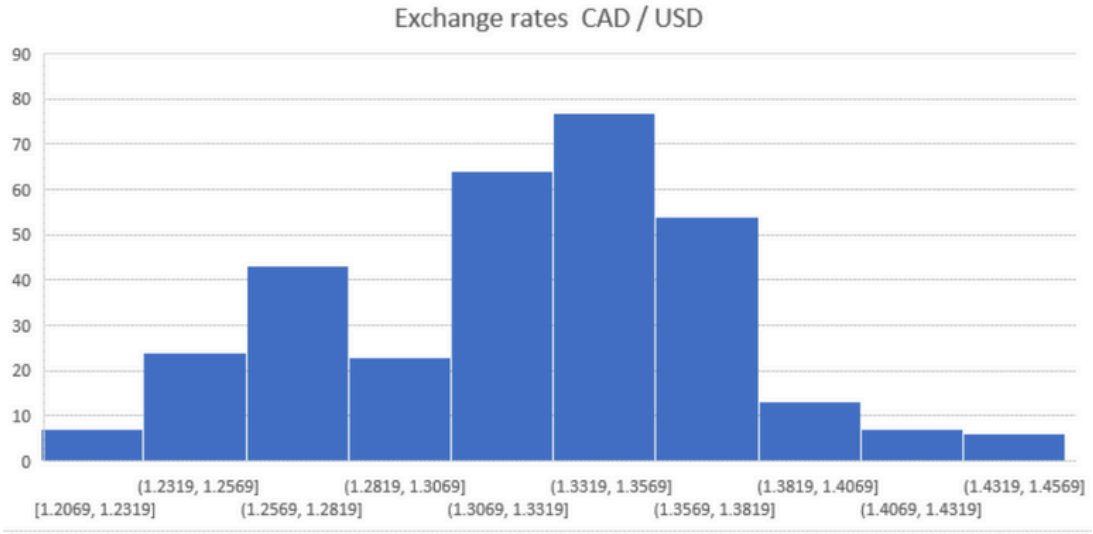


B) Financial Ratios Summary

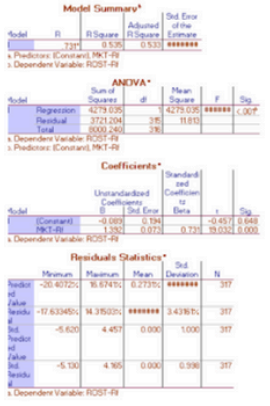
Company	Current Ratio	Debt to Equity Ratio	Return on Assets	Return on Equity	Price to Earnings Ratio
Lockheed Martin	1.25	0.94	0.12	0.40	26.86
The Coca-Cola Company	1.05	2.04	0.096	0.41	27.83
Apple Inc.	1.2	1.33	0.29	1.05	29.17
Tesla Inc.	1.25	0.33	0.07	0.18	71.43
ROSS Stores Inc.	1.33	0.60	0.17	0.42	45.58
Exxon Mobil Corporation	1.29	0.33	0.05	0.14	14.00

Company	Market Price per Share	EPS (USD)	Total Debt (in billions USD)	Total Shareholders' Equity (in billions USD)	Net Income (in billions USD)	Avg. Shareholders' Equity (in billions USD)	Avg. Total Assets (in billions USD)	Current Assets (in billions USD)	Current Liabilities (in billions USD)
Lockheed Martin	$470	$17.50	$15	$16	$6	$15	$50	$ 10.00	$ 8.00
The Coca-Cola Company	$64	$2.30	$47	$23	$9	$22	$94	$ 22.00	$ 21.00
Apple Inc.	$175	$6.00	$120	$90	$100	$95	$350	$ 150.00	$ 125.00
Tesla Inc.	$250	$3.50	$10	$30	$5	$28	$70	$ 25.00	$ 20.00
ROSS Stores Inc.	$145.87	$3.20	$1.50	$2.50	$1.00	$2.40	$6.00	$ 2.00	$ 1.50
Exxon Mobil Corporation	$105	$7.50	$50	$150	$20	$140	$400	$ 90.00	$ 70.00

CAD vs. USD

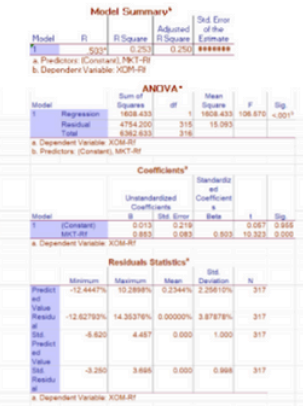


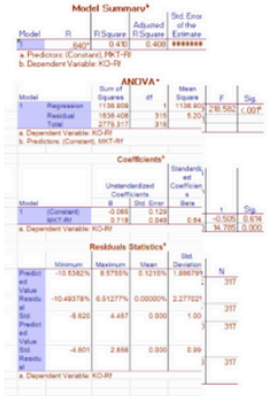
C) CAPM table from SPSS

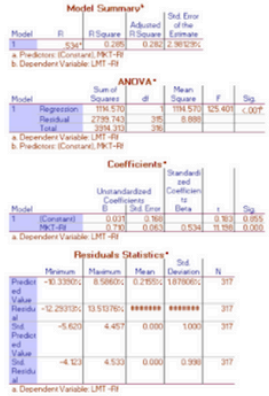


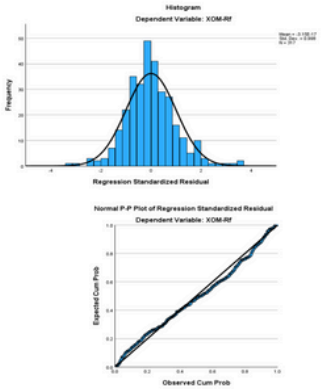


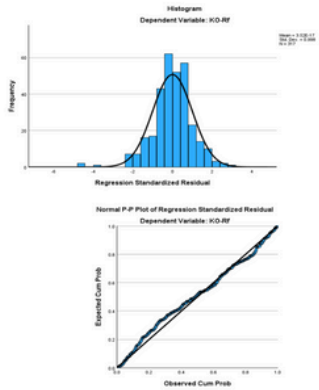


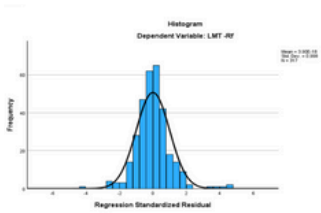


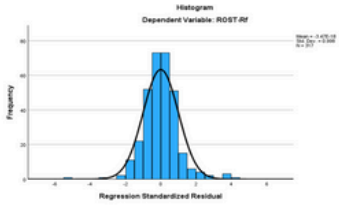


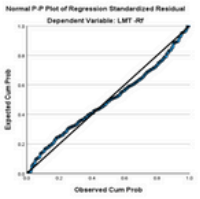


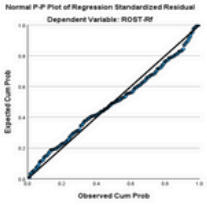


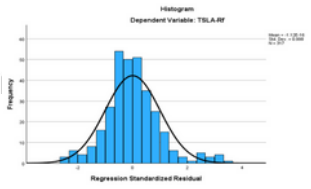


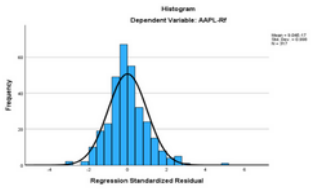


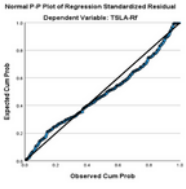


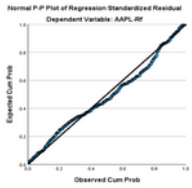












D) FF3 table from SPSS

Multivariate Tests[a]

Effect		Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
Intercept	Pillai's Trace	0.023	1.214[b]	6.000	******	0.299	0.023
	Wilks' Lambda	0.977	1.214[b]	6.000	******	0.299	0.023
	Hotelling's Trace	0.024	1.214[b]	6.000	******	0.299	0.023
	Roy's Largest	0.024	1.214[b]	6.000	******	0.299	0.023
MKTRf	Pillai's Trace	0.844	277.102[b]	6.000	******	0.000	0.844
	Wilks' Lambda	0.156	277.102[b]	6.000	******	0.000	0.844
	Hotelling's Trace	5.398	277.102[b]	6.000	******	0.000	0.844
	Roy's Largest	5.398	277.102[b]	6.000	******	0.000	0.844
SMB	Pillai's Trace	0.119	6.930[b]	6.000	******	0.000	0.119
	Wilks' Lambda	0.881	6.930[b]	6.000	******	0.000	0.119
	Hotelling's Trace	0.135	6.930[b]	6.000	******	0.000	0.119
	Roy's Largest	0.135	6.930[b]	6.000	******	0.000	0.119
HML	Pillai's Trace	0.475	46.433[b]	6.000	******	0.000	0.475
	Wilks' Lambda	0.525	46.433[b]	6.000	******	0.000	0.475
	Hotelling's Trace	0.905	46.433[b]	6.000	******	0.000	0.475
	Roy's Largest	0.905	46.433[b]	6.000	******	0.000	0.475

a. Design: Intercept + MKTRf + SMB + HML
b. Exact statistic

Tests of Between-Subjects Effects

Source		Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	ROST-Rf	4394.000[a]	3	1464.667	127.124	0.000	0.549
	TSLA-Rf	9334.457[b]	3	3111.484	56.661	0.000	0.352
	AAPL-Rf	2978.307[c]	3	992.769	175.463	0.000	0.627
	XOM-Rf	3411.135[d]	3	1137.045	120.581	0.000	0.536
	KO-Rf	1347.850[e]	3	449.283	98.514	0.000	0.486
	LMT -Rf	1388.737[f]	3	462.912	57.370	0.000	0.355
Intercept	ROST-Rf	1.866	1	1.866	0.162	0.688	0.001
	TSLA-Rf	202.686	1	202.686	3.631	0.056	0.012
	AAPL-Rf	22.238	1	22.238	3.930	0.048	0.012
	XOM-Rf	1.196	1	1.196	0.127	0.722	0.000
	KO-Rf	2.048	1	2.048	0.449	0.503	0.001
	LMT -Rf	0.110	1	0.110	0.014	0.907	0.000
MKTRf	ROST-Rf	4052.486	1	*******	351.732	0.000	0.529
	TSLA-Rf	7069.214	1	7069.214	126.733	0.000	0.291
	AAPL-Rf	2711.972	1	2711.972	479.318	0.000	0.605
	XOM-Rf	1459.844	1	1459.844	154.813	0.000	0.331
	KO-Rf	1210.435	1	1210.435	265.411	0.000	0.459
	LMT -Rf	1180.398	1	1180.398	146.269	0.000	0.319
SMB	ROST-Rf	0.177	1	0.177	0.015	0.901	0.000
	TSLA-Rf	293.043	1	293.043	5.336	0.022	0.017
	AAPL-Rf	72.490	1	72.490	12.812	0.000	0.039
	XOM-Rf	8.471	1	8.471	0.858	0.344	0.003
	KO-Rf	80.824	1	80.824	17.722	0.000	0.054
	LMT -Rf	73.524	1	73.524	9.112	0.003	0.028
HML	ROST-Rf	114.615	1	114.615	9.948	0.002	0.031
	TSLA-Rf	933.454	1	933.454	18.091	0.000	0.055
	AAPL-Rf	264.436	1	264.436	46.749	0.000	0.130
	XOM-Rf	1763.657	1	1763.657	183.610	0.000	0.377
	KO-Rf	133.366	1	133.366	29.243	0.000	0.085
	LMT -Rf	204.475	1	204.475	25.341	0.000	0.075
Error	ROST-Rf	3606.240	313	11.522			
	TSLA-Rf	17189.056	313	54.914			
	AAPL-Rf	1770.943	313	5.658			
	XOM-Rf	2351.436	313	3.430			
	KO-Rf	1427.467	313	4.561			
	LMT -Rf	2525.576	313	8.069			
Total	ROST-Rf	8023.885	317				
	TSLA-Rf	27042.165	317				
	AAPL-Rf	4860.262	317				
	XOM-Rf	6380.045	317				
	KO-Rf	2780.000	317				
	LMT -Rf	3923.032	317				
Corrected Total	ROST-Rf	8000.240	316				
	TSLA-Rf	********	316				
	AAPL-Rf	4749.256	316				
	XOM-Rf	6362.633	316				
	KO-Rf	2775.317	316				
	LMT -Rf	3914.313	316				

a. R Squared = .549 (Adjusted R Squared = .545)
b. R Squared = .352 (Adjusted R Squared = .346)
c. R Squared = .627 (Adjusted R Squared = .624)
d. R Squared = .536 (Adjusted R Squared = .532)
e. R Squared = .486 (Adjusted R Squared = .481)
f. R Squared = .355 (Adjusted R Squared = .349)

Parameter Estimates

Dependent Variable		B	Std. Error	t	Sig.	95% Confidence Interval Lower Bound	95% Confidence Interval Upper Bound	Partial Eta Squared
ROST-Rf	Intercept	-0.077	0.192	-0.402	0.688	-0.455	0.300	0.001
	MKTRf	1.389	0.074	18.755	0.000	1.243	1.534	0.529
	SMB	0.015	0.124	0.124	0.901	-0.229	0.260	0.000
	HML	0.247	0.078	3.154	0.002	0.093	0.402	0.031
TSLA-Rf	Intercept	0.805	0.419	1.921	0.056	-0.019	1.629	0.012
	MKTRf	1.834	0.162	11.346	0.000	1.516	2.152	0.291
	SMB	0.628	0.272	2.310	0.022	0.093	1.162	0.017
	HML	-0.728	0.171	-4.253	0.000	-1.065	-0.391	0.055
AAPL-Rf	Intercept	0.267	0.134	1.983	0.048	0.002	0.531	0.012
	MKTRf	1.136	0.052	21.893	0.000	1.034	1.238	0.605
	SMB	-0.312	0.087	-3.579	0.000	-0.484	-0.141	0.039
	HML	-0.376	0.055	-6.837	0.000	-0.484	-0.268	0.130
XOM-Rf	Intercept	0.062	0.174	0.356	0.722	-0.280	0.403	0.000
	MKTRf	0.833	0.067	12.442	0.000	0.702	0.965	0.331
	SMB	0.107	0.113	0.948	0.344	-0.115	0.328	0.003
	HML	0.978	0.071	13.777	0.000	0.838	1.117	0.377
KO-Rf	Intercept	-0.081	0.121	-0.670	0.503	-0.318	0.157	0.001
	MKTRf	0.759	0.047	16.291	0.000	0.667	0.851	0.459
	SMB	-0.330	0.078	-4.210	0.000	-0.484	-0.176	0.054
	HML	0.267	0.049	5.408	0.000	0.170	0.364	0.085
LMT -Rf	Intercept	0.079	0.161	0.117	0.907	-0.237	0.335	0.000
	MKTRf	0.749	0.062	12.095	0.000	0.627	0.871	0.319
	SMB	-0.314	0.104	-3.019	0.003	-0.519	-0.109	0.028
	HML	0.330	0.066	5.034	0.000	0.201	0.460	0.075

E) Portfolio Optimization

Var/Covar Matrix

	ROST-Mean	TSLA-Mean	AAPL-Mean	XOM-Mean	KO-Mean	LMT -Mean
ROST-Mean	0.002149	0.001184	0.000829	0.000866	0.000741	0.000681
TSLA-Mean	0.001184	0.009272	0.001775	0.000346	0.000299	0.000317
AAPL-Mean	0.000829	0.001775	0.001581	0.000326	0.000371	0.000347
XOM-Mean	0.000866	0.000346	0.000326	0.0019	0.000496	0.000739
KO-Mean	0.000741	0.000299	0.000371	0.000496	0.000761	0.000538
LMT -Mean	0.000681	0.000317	0.000347	0.000739	0.000538	0.001251

Efficient Frontier Table:

St. Dev	ROST	TSLA	AAPL	XOM	KO	LMT	Total	Expected Return	Expected St.Dev
2.49%	0.00%	0.01%	21.29%	8.96%	54.89%	14.84%	100.00%	0.32%	2.49%
2.74%	0.00%	0.01%	48.92%	12.08%	22.93%	16.05%	100.00%	0.46%	2.74%
2.99%	0.00%	0.01%	61.42%	13.39%	8.53%	16.64%	100.00%	0.53%	2.99%
3.24%	0.00%	0.01%	72.31%	13.54%	0.00%	14.14%	100.00%	0.58%	3.24%
3.49%	0.00%	0.01%	82.79%	11.81%	0.00%	5.39%	100.00%	0.62%	3.49%
3.72%	0.00%	17.21%	59.71%	12.45%	0.00%	10.64%	100.00%	0.71%	3.72%
3.74%	0.00%	16.06%	63.53%	11.90%	0.00%	8.51%	100.00%	0.71%	3.74%
3.99%	0.00%	20.21%	63.19%	11.28%	0.00%	5.33%	100.00%	0.76%	3.99%
4.24%	0.00%	22.82%	66.20%	10.37%	0.00%	0.62%	100.00%	0.79%	4.24%
4.49%	0.00%	26.00%	67.43%	5.97%	0.00%	0.60%	100.00%	0.83%	4.49%

Performance Metric Calculations

For estimating Alpha and Beta, we calculated portfolio returns for the time period by multiplying the weights of each portfolio by the return of the assets and adding them up for each week. Then, we subtracted the risk-free rate from them, and regressed them against Mkt-Rf. Here are the regression results:

SUMMARY OUTPUT

Regression Statistics	
Multiple F	0.787889
R Square	0.620769
Adjusted I	0.619565
Standard E	0.0154
Observati	317

ANOVA

	df	*SS*	*MS*	*F*	*gnificance F*
Regressio	1	0.122285	0.122285	515.6279	2.71E-68
Residual	315	0.074704	0.000237		
Total	316	0.196989			

	Coefficients	tandard Err	t Stat	P-value	Lower 95%	Upper 95%	ower 95.0%	pper 95.0%
Intercept	0.000797	0.000869	0.917057	0.359814	-0.00091	0.002507	-0.00091	0.002507
Mkt-Rf	0.747327	0.032911	22.70744	2.71E-68	0.682574	0.812081	0.682574	0.812081

SUMMARY OUTPUT

Regression Statistics	
Multiple F	0.821845
R Square	0.67543
Adjusted I	0.674399
Standard E	0.021269
Observati	317

ANOVA

	df	SS	MS	F	gnificance F
Regressio	1	0.296537	0.296537	655.5141	5.86E-79
Residual	315	0.142497	0.000452		
Total	316	0.439034			

	Coefficients	tandard Err	t Stat	P-value	Lower 95%	Upper 95%	ower 95.0%	pper 95.0%
Intercept	0.003587	0.0012	2.988464	0.003024	0.001225	0.005949	0.001225	0.005949
Mkt-Rf	1.163763	0.045454	25.60301	5.86E-79	1.074331	1.253195	1.074331	1.253195

SUMMARY OUTPUT

Regression Statistics	
Multiple F	0.510075
R Square	0.260176
Adjusted I	0.257828
Standard E	0.083085
Observati	317

Plot Area

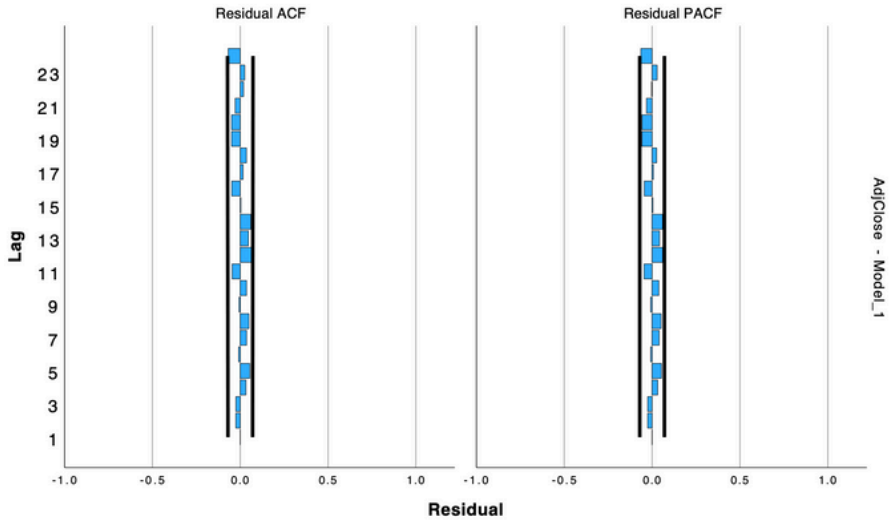
ANOVA

	df	SS	MS	F	gnificance F
Regressio	1	0.764709	0.764709	110.7771	2.13E-22
Residual	315	2.174485	0.006903		
Total	316	2.939194			

	Coefficients	tandard Err	t Stat	P-value	Lower 95%	Upper 95%	ower 95.0%	pper 95.0%
Intercept	0.008178	0.004689	1.744114	0.082115	-0.00105	0.017404	-0.00105	0.017404
Mkt-Rf	1.868844	0.177561	10.52507	2.13E-22	1.519488	2.2182	1.519488	2.2182

We also calculated the cumulative return of each portfolio starting at 100% on Jan 1st, 2019.

E) Residual Plots for ARIMA forecasting



G) ARIMA analysis: Model fit Analysis

Model Fit

Fit Statistic	Mean	SE	Minimum	Maximum	Percentile 5	Percentile 10
Stationary R-squared	7.772E-16	.	7.772E-16	7.772E-16	7.772E-16	7.772E-16
R-squared	.990	.	.990	.990	.990	.990
RMSE	.959	.	.959	.959	.959	.959
MAPE	.493	.	.493	.493	.493	.493
MaxAPE	3.754	.	3.754	3.754	3.754	3.754
MAE	.703	.	.703	.703	.703	.703
MaxAE	5.142	.	5.142	5.142	5.142	5.142
Normalized BIC	-.074	.	-.074	-.074	-.074	-.074

Model Fit

Fit Statistic	Percentile 25	50	75	90	95
Stationary R-squared	7.772E-16	7.772E-16	7.772E-16	7.772E-16	7.772E-16
R-squared	.990	.990	.990	.990	.990
RMSE	.959	.959	.959	.959	.959
MAPE	.493	.493	.493	.493	.493
MaxAPE	3.754	3.754	3.754	3.754	3.754
MAE	.703	.703	.703	.703	.703
MaxAE	5.142	5.142	5.142	5.142	5.142
Normalized BIC	-.074	-.074	-.074	-.074	-.074

H) Daily Return for USD/JPY

95% confidence interval	**-1.09%**	=PERCENTILE(D2:D782,0.05)
99% confidence interval	**-1.73%**	=PERCENTILE(D2:D782,0.01)

